

Hence advanced HMDs must feature a system which allows them to command the system using voice. But of course voice may not always work well in all AR systems. For example, if an architect is viewing a 2D drawing of a building in 3D using a HMD, commanding the HMD with voice to show minor changes in design will be a much more frustrating experience than using a wand or stylus to alter his view.

Another way to send commands to an AR system is in the form a glove which detects the subject's hand gestures and sends them to the processing unit which then takes an appropriate action based on that gesture. Yet another method to send manual input to an AR system would be touch. For example, an AR browser could be designed to show extra information about a building in front only if you touch the image on your mobile phone's display.

Input hardware for tracking changes in environment

Before going further, we want to clear the point that all interaction hardware are input hardware in their own right. They allow the user to input commands to the system. What we are going to talk about in this part is not about them. In this part we talk about the hardware which allow the AR implementations to function automatically (and intuitively). Let's start with the implementation of AR which we can encounter first hand - AR browser applications built for mobile phones.

Use of AR on your mobile phone to assist you during your journey is one of the simplest usage of AR systems. Most AR browsers use a lot of information sources built into your phone to add information to your view of places around you. We have said earlier that AR browsers as such do not understand what they see. They show the information from the server (for the app) which depends on your phone sending its location and orientation. To learn the location, the AR browsers use the GPS module built into your mobile phone. If you are not getting GPS information, then the server for the application cannot know your exact location and AR would either not work at all or if it works, it would be inaccurate. The second piece of information it needs to know is which direction you are facing which is supplied by the compass built into the phone. The third source needed is what angle you are (your phone is) looking at from the ground. For example if you are trekking and you point your phone towards the base camp then it is important for the AR application to know that you were looking down or else it might mark the base camp as the hill's peak. The information about the angle of